

Robot Learning

Lecture 1. Foundation of Sensing and Actuation

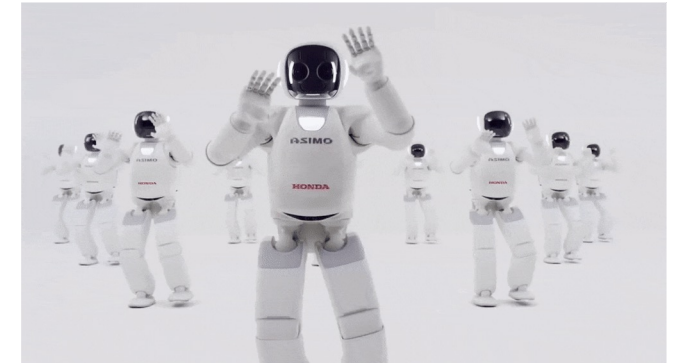
What is a robot

From Encyclopedia Britannica:

robot, any automatically operated machine that replaces human effort, though it may not resemble human beings in appearance or perform functions in a humanlike manner.

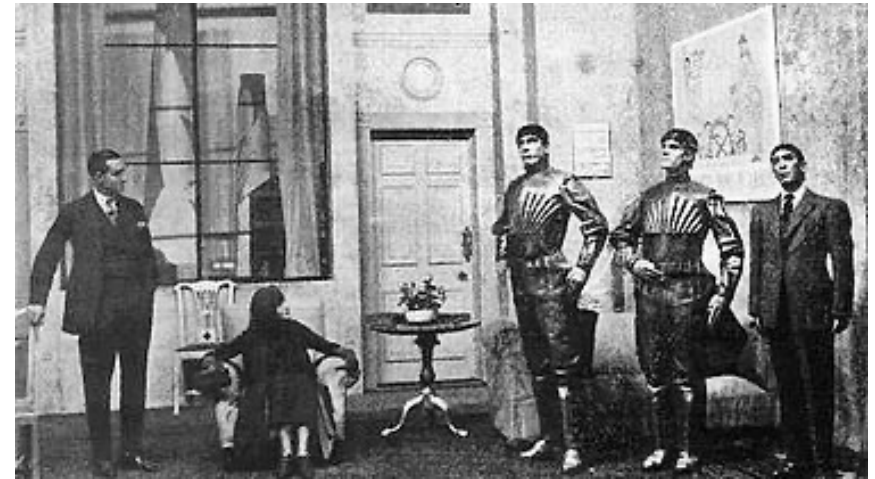


Robot Learning



Historical overview

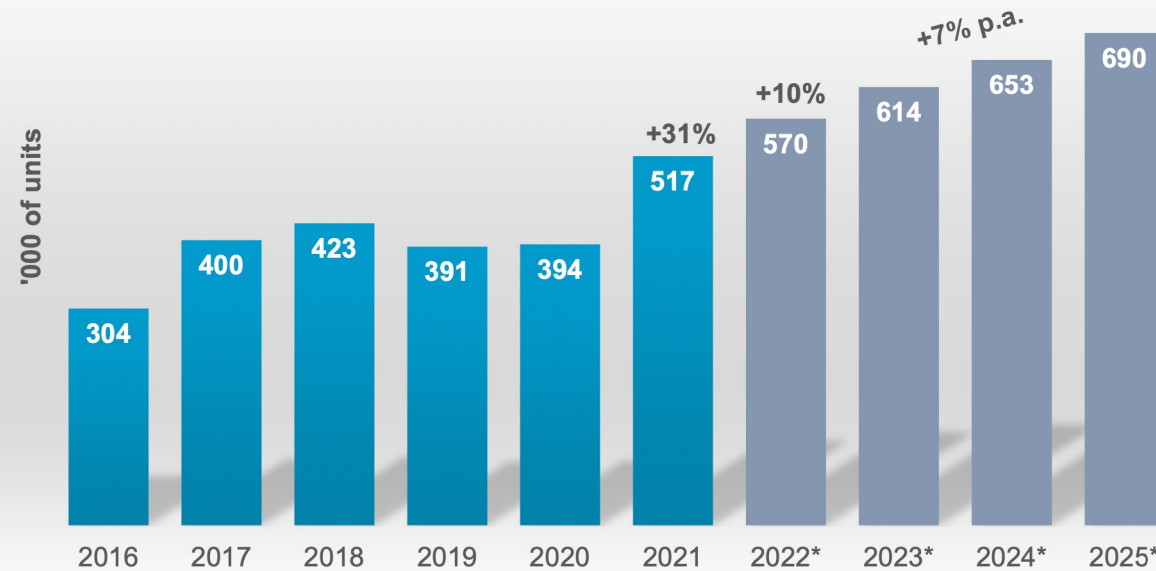
- **Robota** (= “work” in slavic languages) are artificial human-like creatures built for being inexpensive workers in the theater play Rossum’s Universal Robots (R.U.R.) written by Karel Capek in 1920
- **Laws of Robotics** by Isaac Asimov in I, Robot (1950)
 1. **A robot may not injure a human being** or, through inaction, allow a human being to come to harm
 2. **A robot must obey orders given to it by human beings**, except where such orders would conflict with the First Law
 3. **A robot must protect its own existence** as long as such protection does not conflict with the First or Second Law



Robotics in numbers

Excellent growth prospects for 2022

Annual installations of industrial robots 2016-2021 and 2022*-2025*



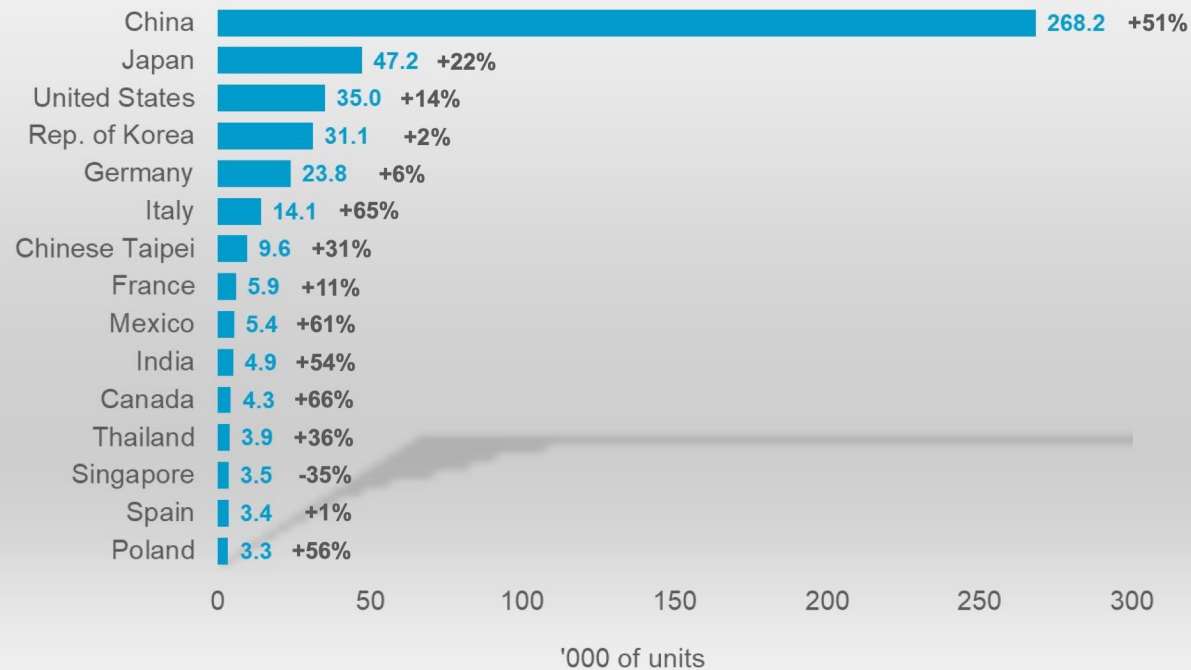
*forecast

Source: World Robotics 2022

Robotics in numbers

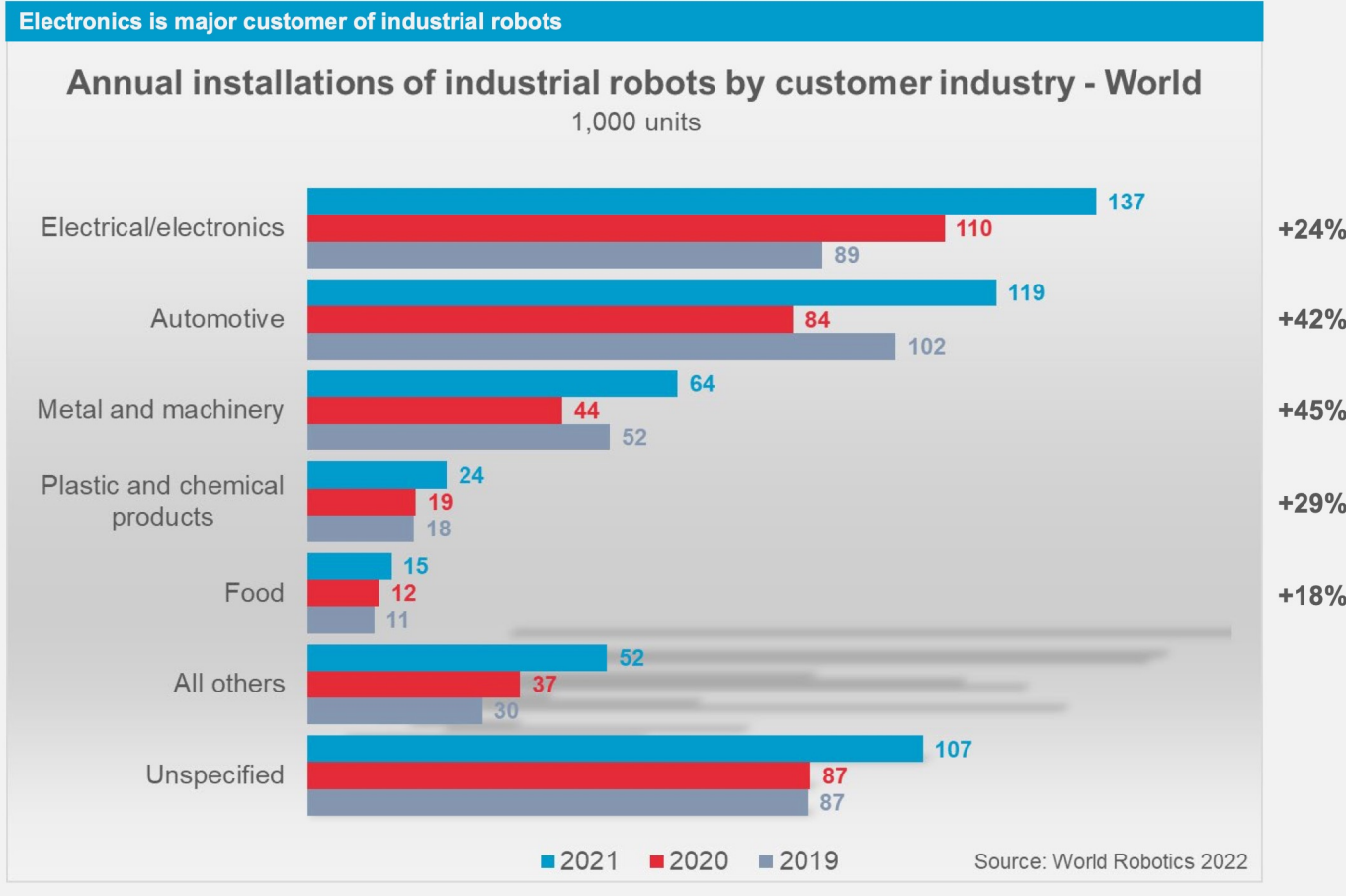
China increases its lead

Annual installations of industrial robots
15 largest markets 2021



Source: World Robotics 2022

Robotics in numbers





Politecnico
di Torino



Robotics in numbers

**24% of global retail sales
online by 2026**

Statista



**15 mn health care
workers missing
worldwide by 2030**

WHO



**Building square
footage +39% by 2050
in the US**

Health Care Weekly



What's next?

**Worldwide housing crisis -
impacting 1.6 billion people
by 2025**

Worldbank

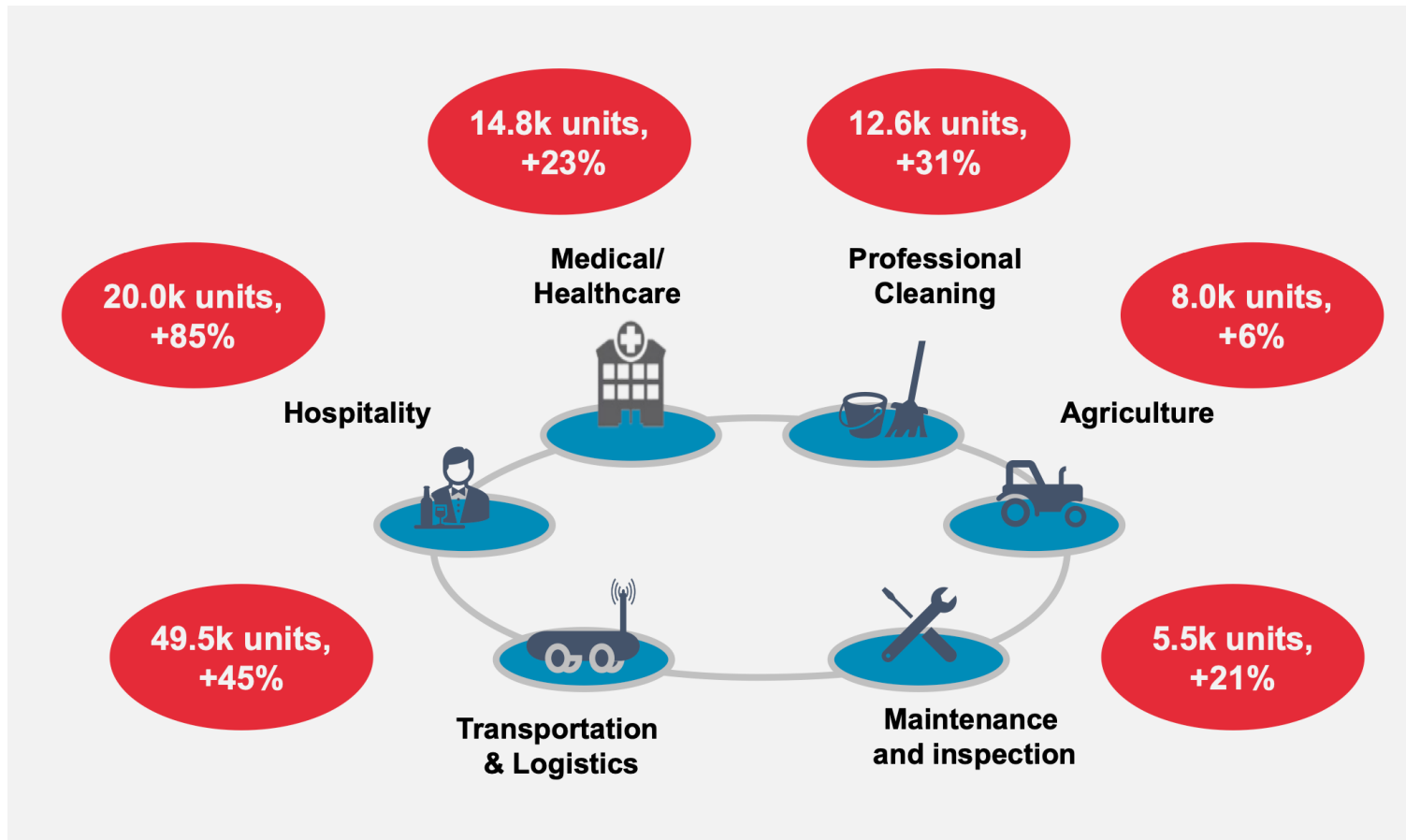


**Glyphosat ban
worldwide**

WHO



Robotics in numbers



Industrial Manipulators



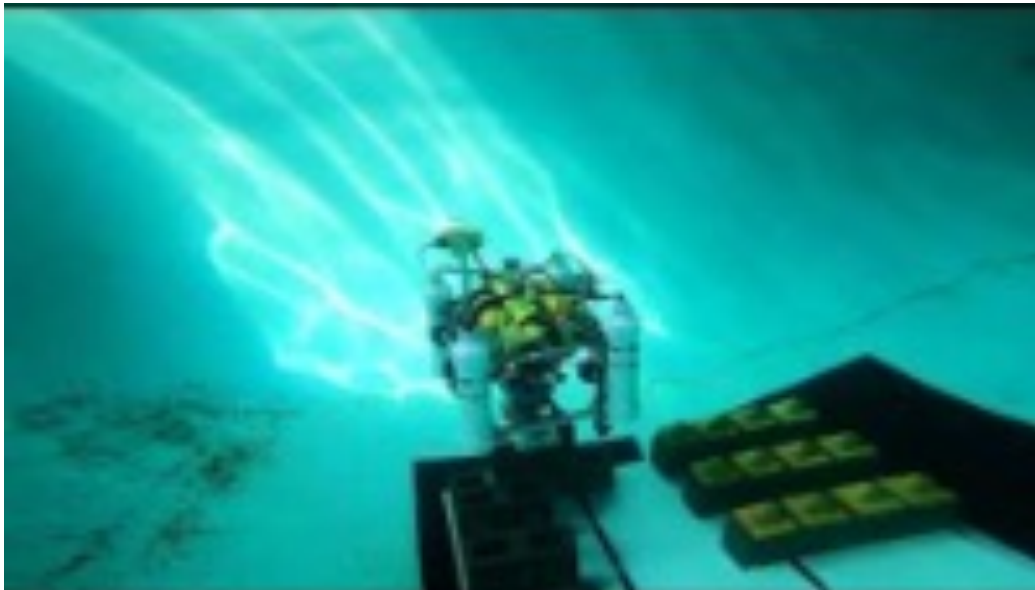
- Handling heavy tasks
- Precision activities
- Repetitive movements

Space Robotics

- Acts on unreachable spots
- Risky environments
- Unusual working conditions



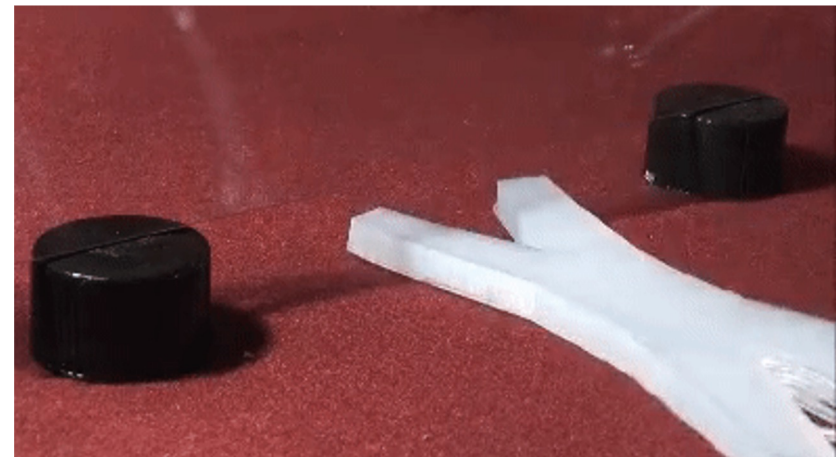
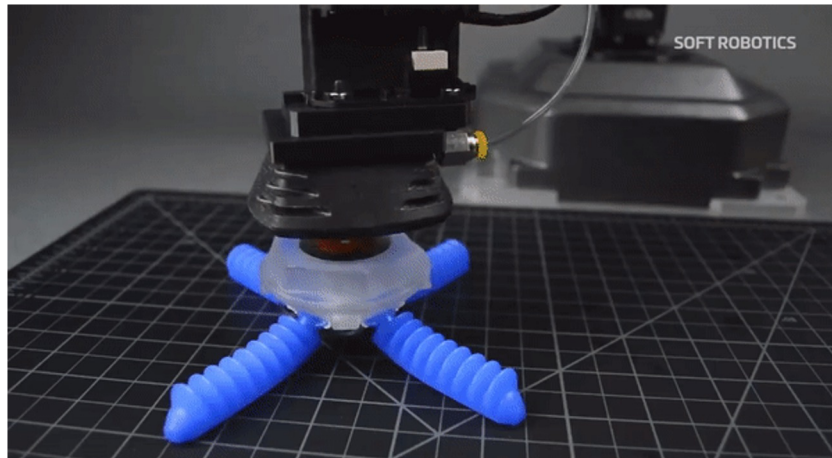
Underwater



- Similar to space applications, but with a lot more water 😊
- Challenging for communication difficulties, high pressure, very low visibility...

Soft Robotics

- New and rapidly growing research area in robotics community
- Use of soft and **elastic** materials (e.g., silicone rubbers)
- Inspiration by biological bodies, motor capabilities of animals
- Intrinsic **flexibility** and **adaptability** to the surrounding environment



Autonomous vehicles

- Very low latency
- Robustness to external noise, modeling error



Champion-Level Performance in Drone Racing using Deep Reinforcement Learning

E. Kaufmann, L. Bauersfeld, A. Loquercio, M. Müller, V. Koltun, D. Scaramuzza



University of
Zurich UZH



- Need to “understand” the surrounding env.
- Risk evaluation and mitigation
- And of course, planning and control of non-linear system

Assistive robotics & Exos.



- Pay attention to exchanged forces
- Safety, security
- Handle unstructured environments

- Everything more complicated because of strict interaction with humans



Active prosthesis

- A particular case of assistive robotics
- Weight and usability is crucial
- Need to simplify the control for an easy interaction



Everyday robotics



- Learn procedural activities
- Learn to cooperate

- Be aware of humans around
- Learn to use tools made for humans



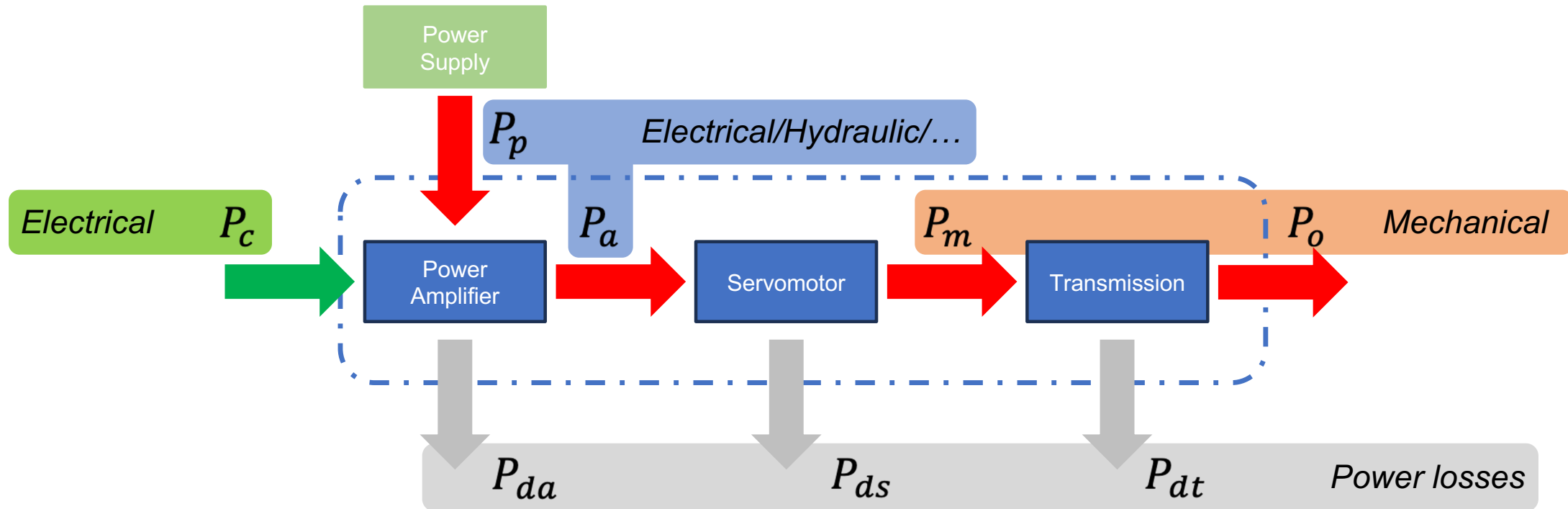
Shopping List

- **Actuators:** electro-mechanical systems which **generate forces** on links, objects, environment
- **Sensors:** electro-mechanical (but also chemical, ...) which provide a **measurable output** proportional to **changes** of a specific physical phenomenon
- **Control unit:** micro-controller, CPU, FPGA, which **regulates** the **actuators** depending on **sensors** readings

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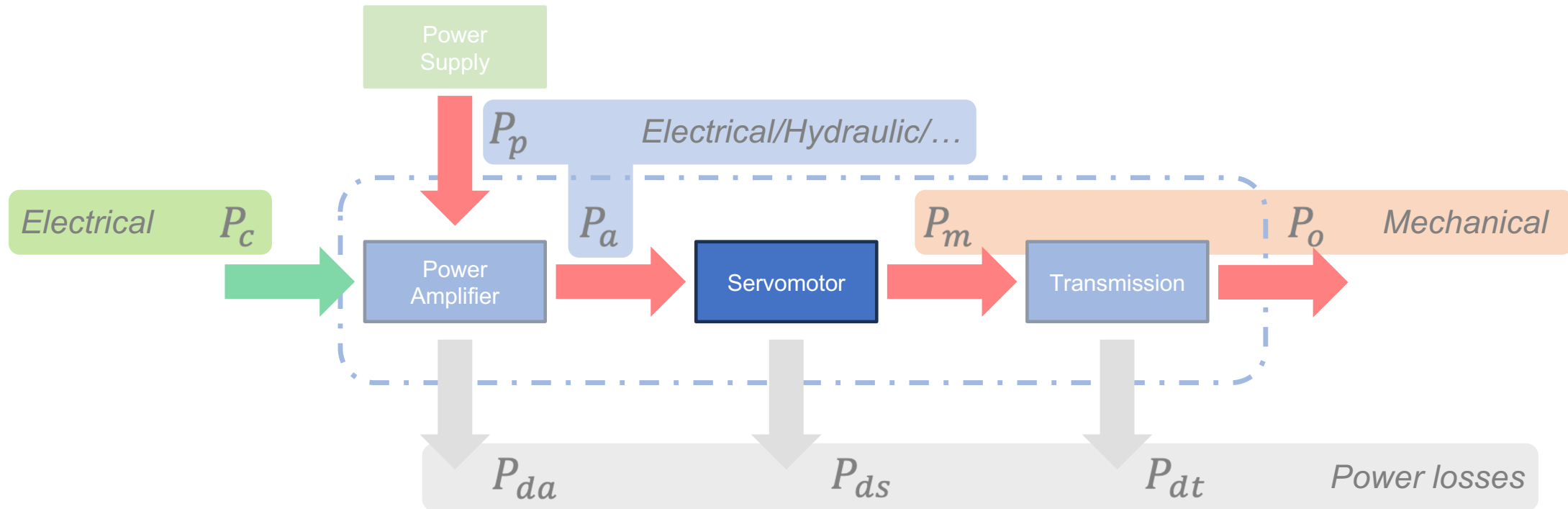
Actuation pipeline



power = force $\times$ speed = torque $\times$ angular speed [Nm/s, W]

efficiency = power out / power in [%]

Actuation pipeline



power = force $\times$ speed = torque $\times$ angular speed [Nm/s, W]

efficiency = power out / power in [%]

Desiderata for Servos in robotics

- low inertia
- high power-to-weight ratio
- high acceleration capabilities
 - variable motion regime, with several stops and inversions
- large range of operational velocities
 - 1 to 1000 turns/min
- high accuracy in positioning
 - at least 1/1000 of a turn
- low torque ripple
 - continuous rotation at low speed
- power: 10W to 10 kW

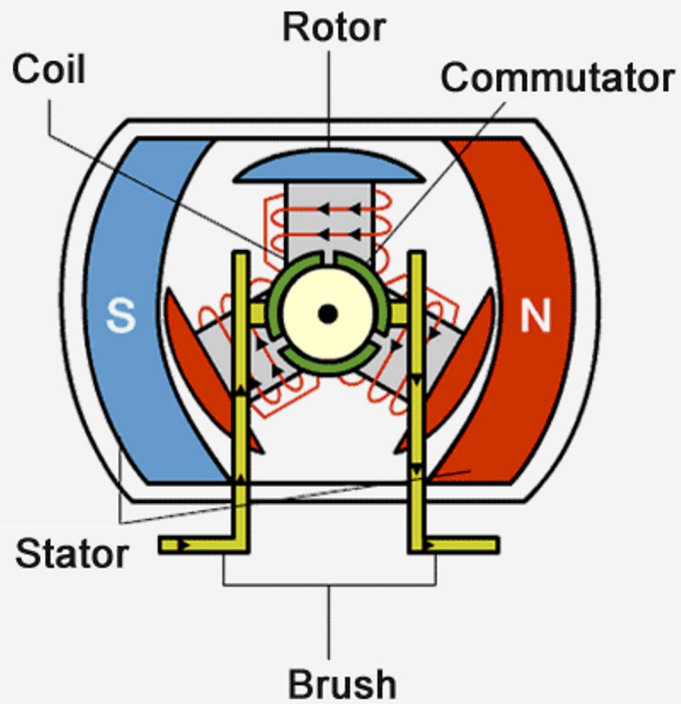
Type of Servos

- Pneumatic/Hydraulic: use of fluid-based pneumatic/hydraulic energy (through pistons, compressors, energy tanks)
 - Usually good for high-force applications
 - Often used to model and replicate artificial muscles
 - Excellent power-to-weight ratio
- Hard to control
- Need external supply
- High cost/maintenance burden
- Poor power efficiency conversion

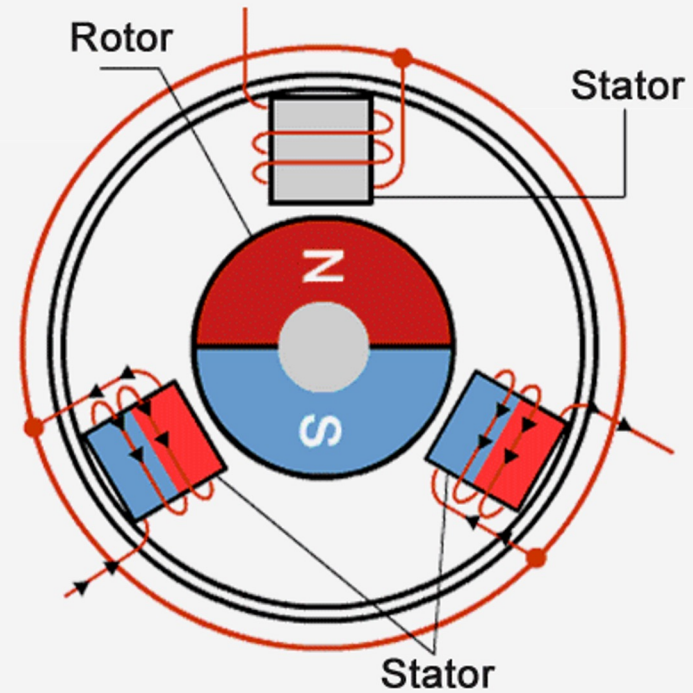
Type of Servos

- Electric: use electro-magnetic energy
 - Power supply easily accessible
 - Many different solutions (even low cost)
 - Better power conversion efficiency
 - Easy maintenance
- Always powered on (e.g. to compensate gravity) – overheating risk

Electrical servos



DC Brush Motor

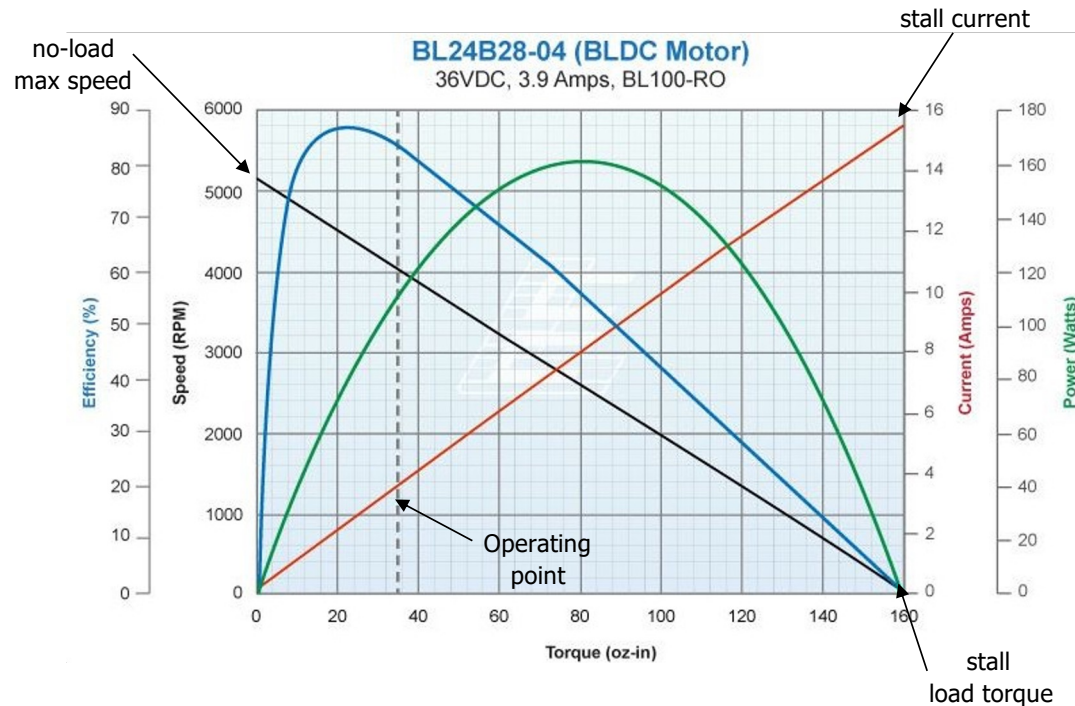


DC Brushless Motor

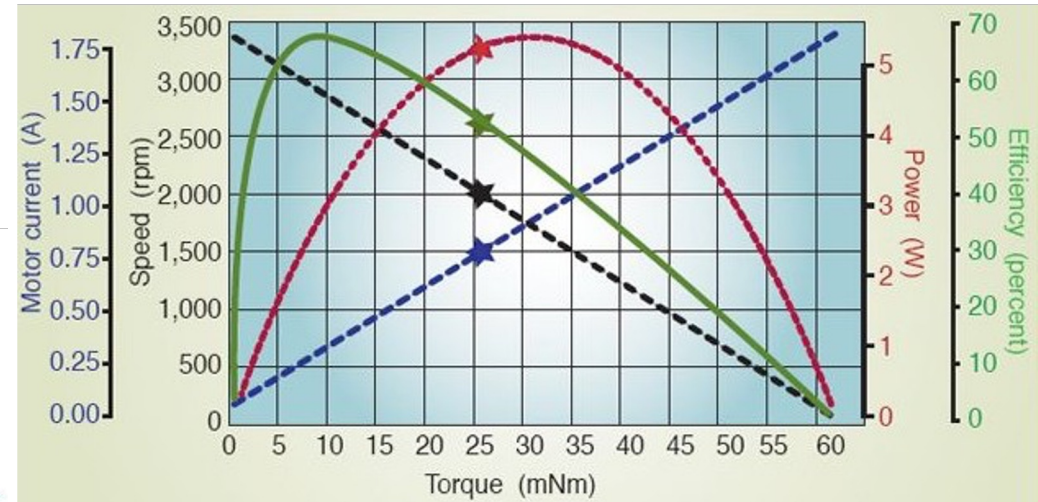
Brushless motors

- Minimum power loss (no friction, no contacts w. brushes)
- Easy to maintain (no brushes replacement)
- Simpler mechanics, more compact

Characteristic curves

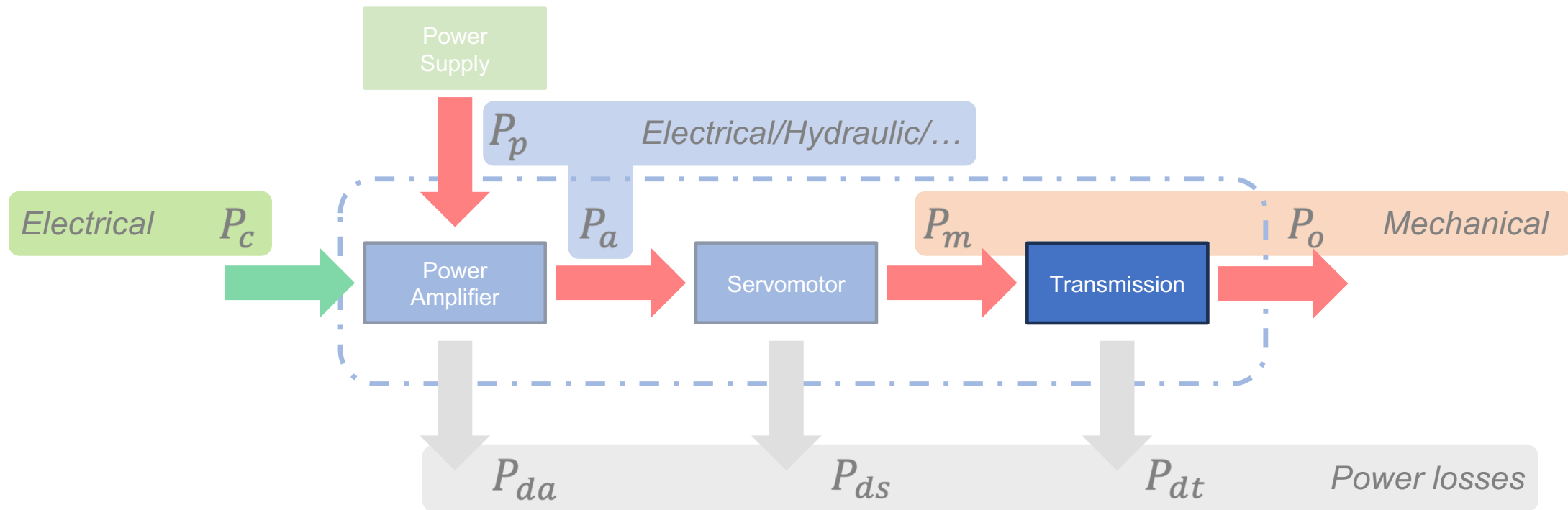


large motor 160W



small motor 5.5W

Actuation pipeline



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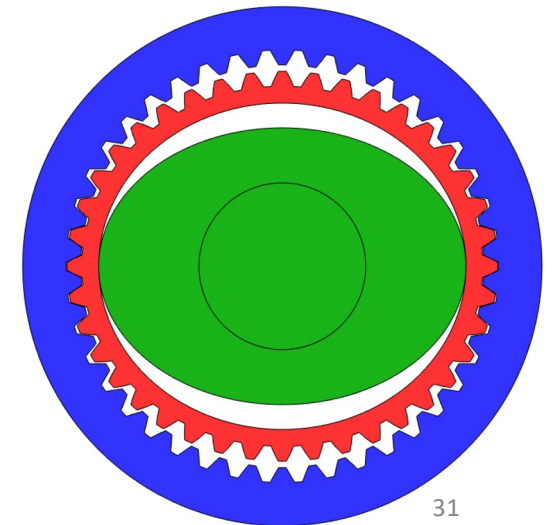
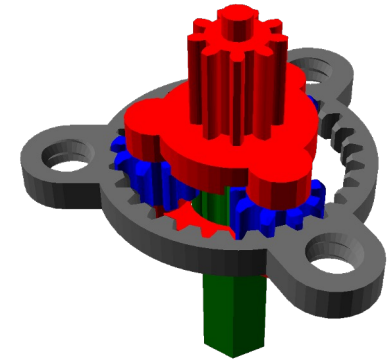


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Actuation pipeline



Transmission



A.Y. 2025-2026

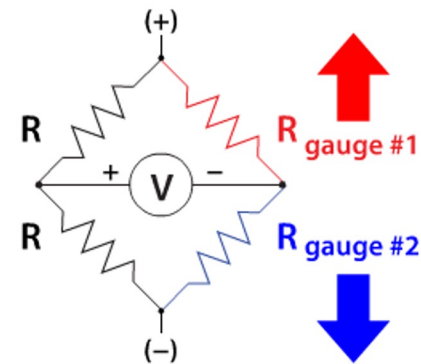
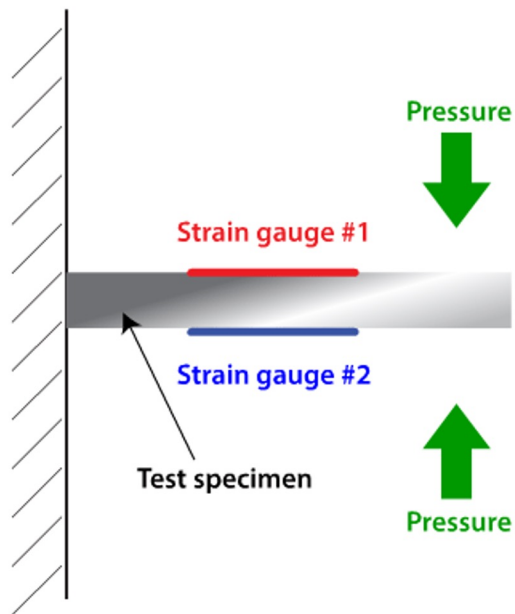
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What is a sensing system

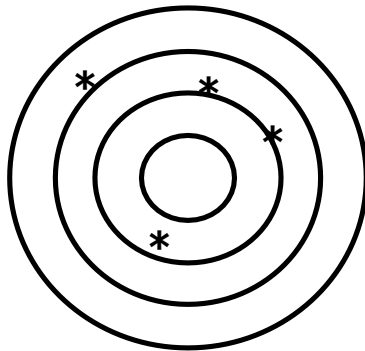
- Convert a physical phenomenon in electric signal (or other measurable quantity)



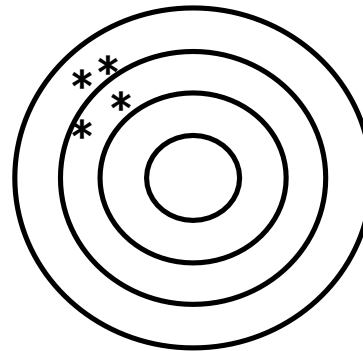
Properties

- **Accuracy:** "precision" of the measured value w.r.t. a given reference (e.g. difference with the correct value)
- **Repeatability:** capability to produce similar outputs when replicating the same input quantity
- **Stability:** capability to keep the same measure over time (or temperature,...)

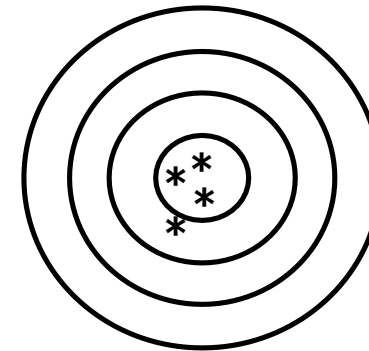
Accuracy vs. Repeatability



low accuracy
low repeatability



low accuracy
high repeatability



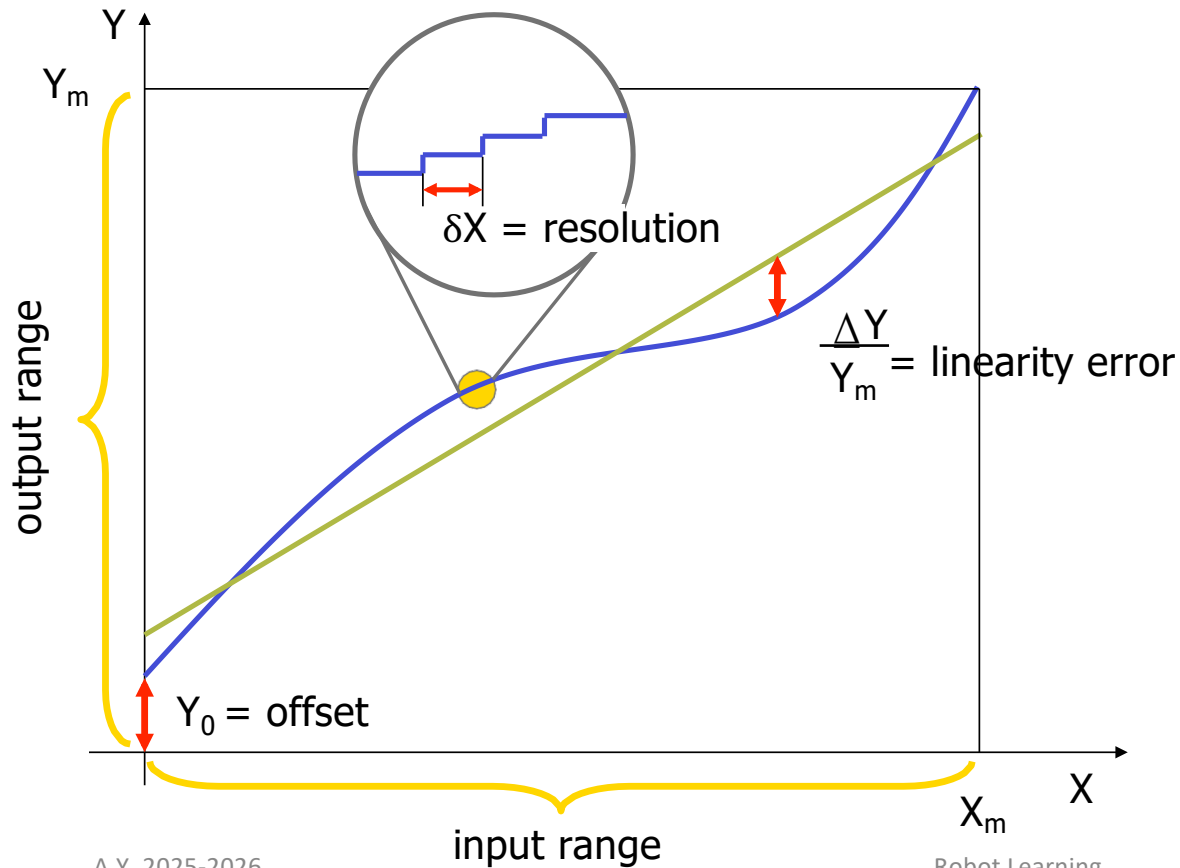
high accuracy
high repeatability

Accuracy vs. Repeatability in robotics



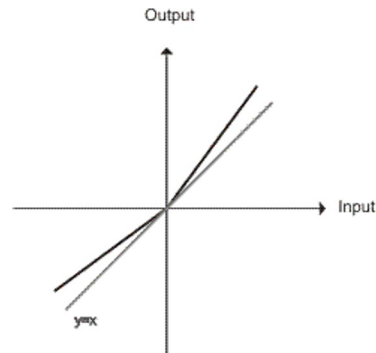
- **Accuracy** and **Repeatability** can vary in different areas of the workspace
- **Standards** (ISO 9283) to define conditions for robot performance assessment

Properties of a sensor

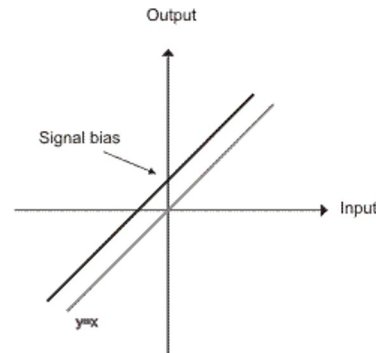


- **Linearity error:** max deviation from linear characteristics
- **Resolution error:** max variation of input which does not produce change in output
- **Offset error:** value measured at zero input

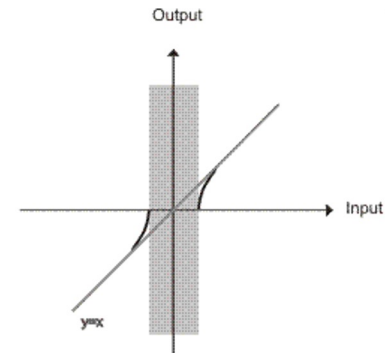
Properties of a sensor



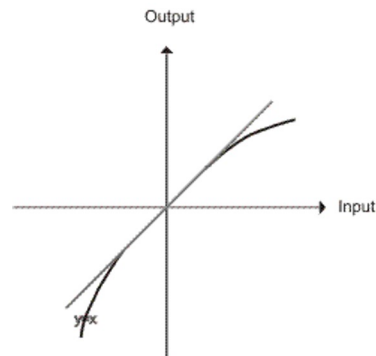
Asymmetry



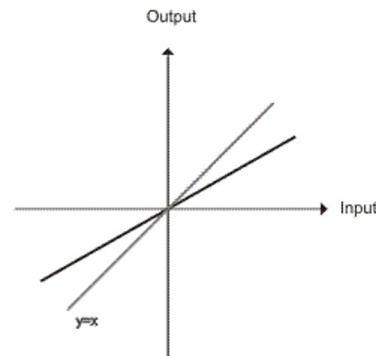
Bias



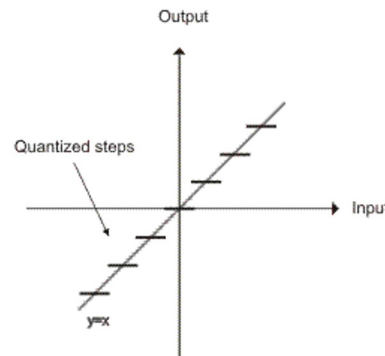
Dead zone



Nonlinearity



Scaling factor



Quantization

Types of sensing

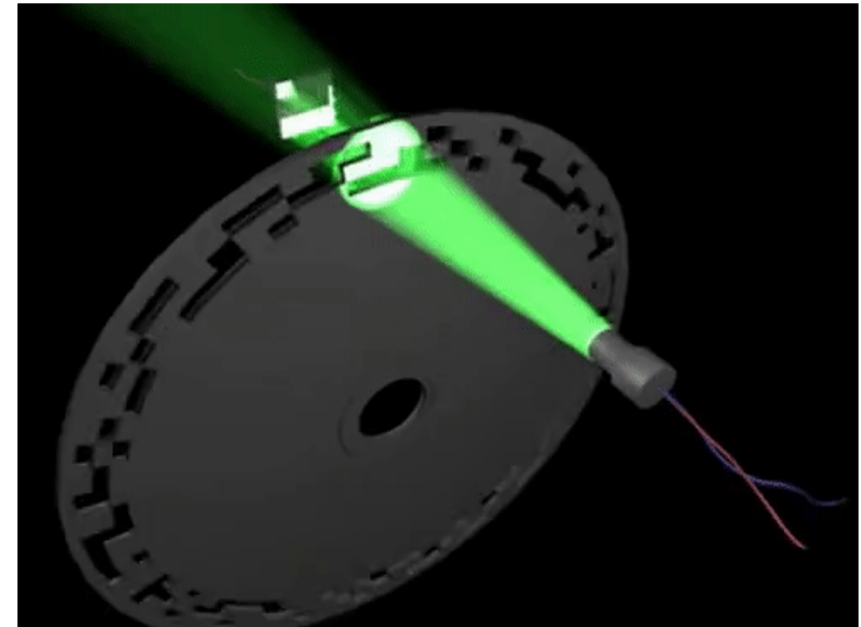
- **Proprioception:** perception of robot own state
 - position of links
 - velocities, accelerations
 - forces
- **Exteroception:** perception of quantities related to external world
 - vision, sound
 - contact forces
 - proximity

Proprioception: **Position**

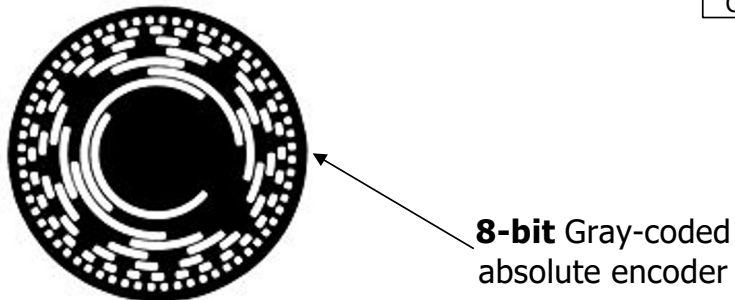
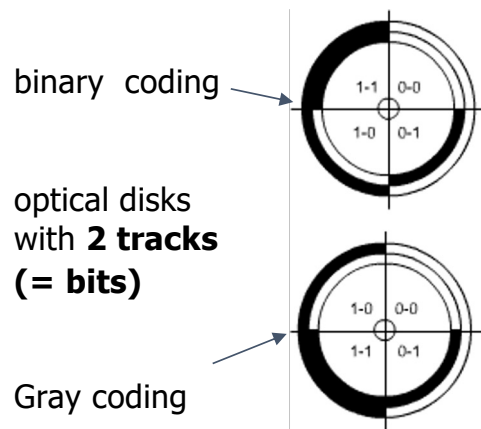
- **Measure the robot current pose:** electrical signal proportional to the displacement of a link w.r.t. a reference (angular, linear)
 - potentiometers, extensometers (linear)
 - resolvers, encoders (angular)
- All the manipulators available on the market are endowed with sensors to keep track of the current angular status of joints
 - **Encoders** are by far the most common

Encoders

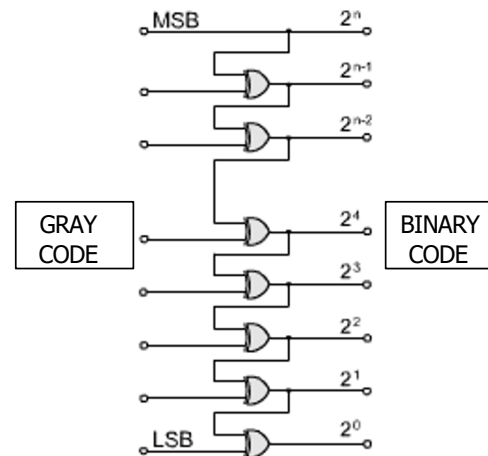
- Usually a **rotating disk**, composed by alternated dark and transparent parts
- A **light beam** is **emitted** from one side of the disk and **detected** by a receiver on the other side (and converted to an electrical signal)
- **Transitions from dark to transparent spots depend on the rotation of the disk** → by properly measuring low-high signal transitions it is possible to infer the current angular position of the disk



Absolute measure



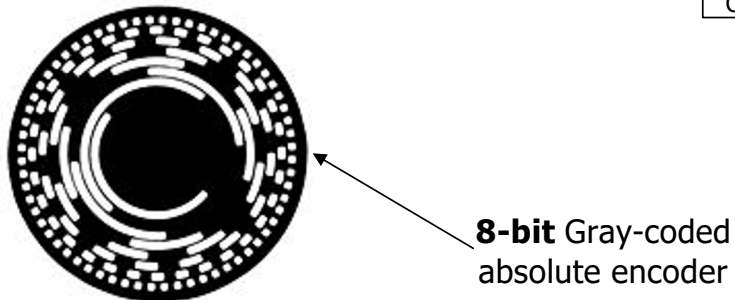
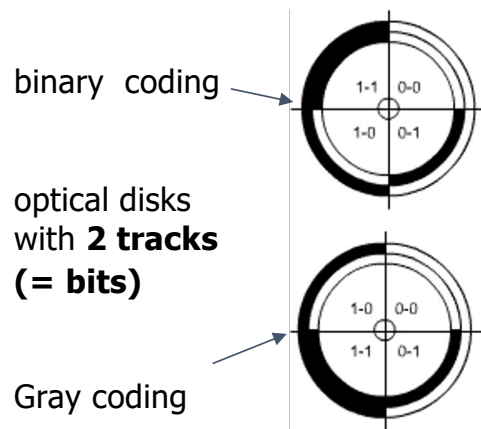
XOR



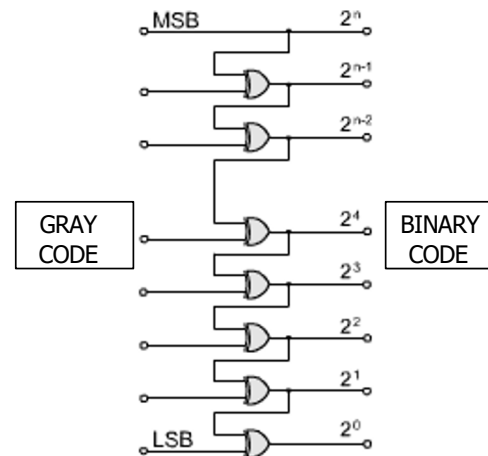
DECIMAL	BINARY	GRAY
0	0000	0000
1	0001	0001
2	0010	0011
3	0011	0010
4	0100	0110
5	0101	0111
6	0110	0101
7	0111	0100
8	1000	1100
9	1001	1101
10	1010	1111
11	1011	1110
12	1100	1010
13	1101	1011
14	1110	1001
15	1111	1000

• Resolution: $360/2^{N_{tracks}}$

Absolute measure



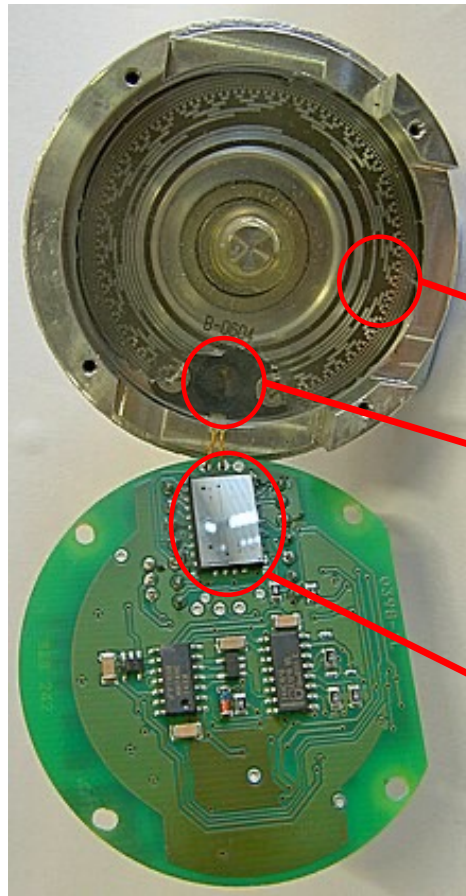
XOR



DECIMAL	BINARY	GRAY
0	0000	0000
1	0001	0001
2	0010	0011
3	0011	0010
4	0100	0110
5	0101	0111
6	0110	0101
7	0111	0100
8	1000	1100
9	1001	1101
10	1010	1111
11	1011	1110
12	1100	1010
13	1101	1011
14	1110	1001
15	1111	1000

• Resolution: $360/2^{N_{tracks}}$
 A.Y. 2025-2026

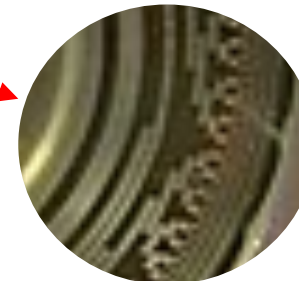
Absolute measure



- Example of gray code absolute rotary encoder

- 13 tracks
- $2^{13} = 8192$ steps per revolution (resolution = 0.044°)

Light emitter

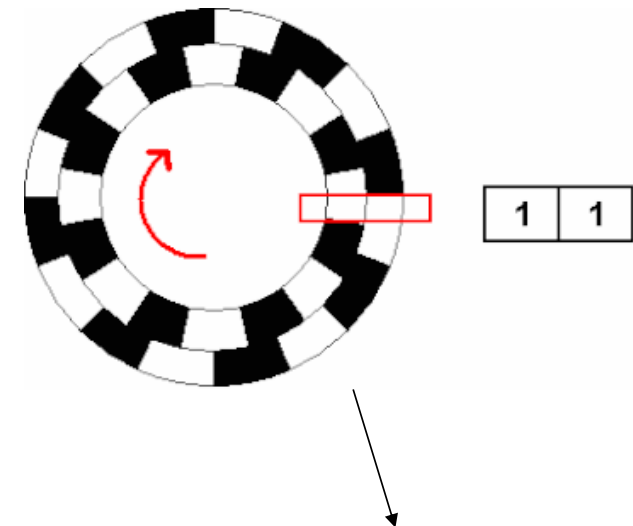


Tracks

Receiver

Relative measure

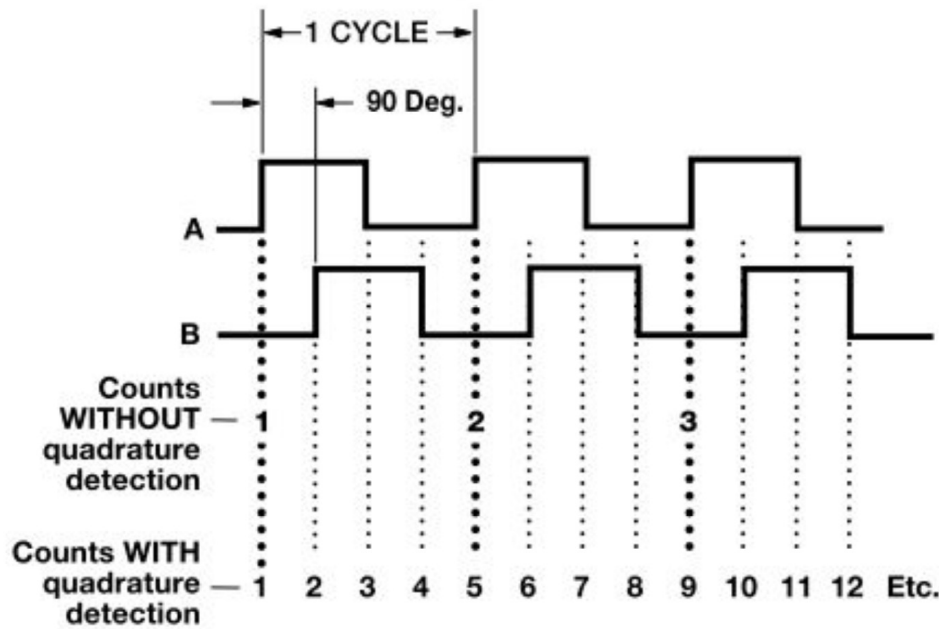
- Also in this case, a rotating disk with alternated dark and transparent parts
- With only two (three) tracks, it is possible to detect **incremental** angular changes
- Two tracks (A, B) are 90° time-shifted (quadrature) to enable detection of rotation direction
- A third channel is used to provide a reference (zero)
- **Resolution** $4N_e$



N_e number of pulses per track
defines the **resolution** ($10^2 - 10^4$)

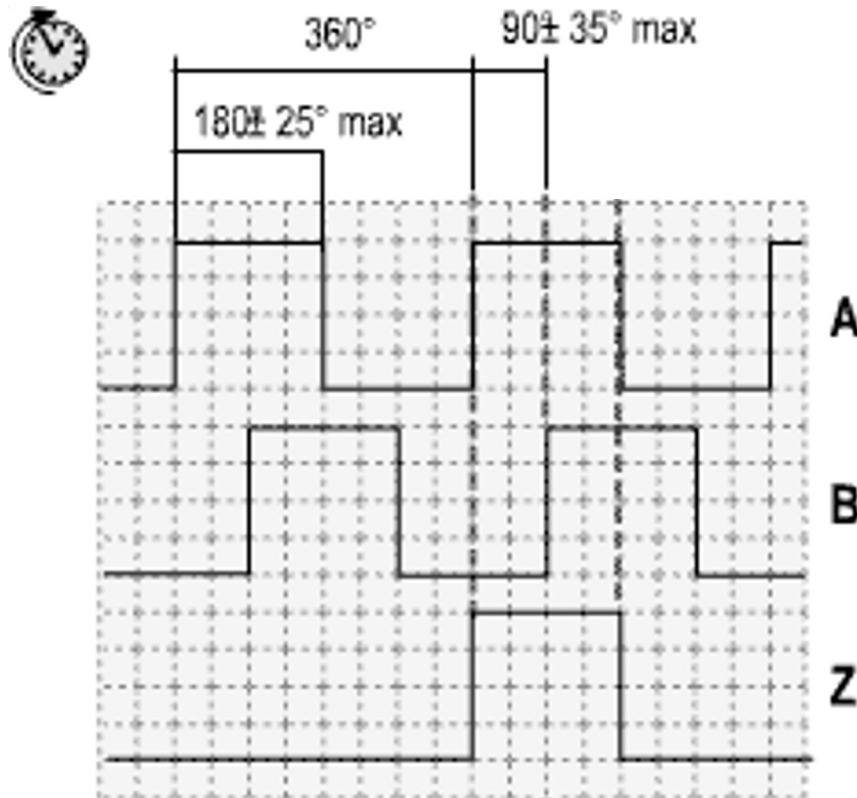
In this example $N_e = 8$

Relative measure



- Works by **counting the transitions** between states
- Only provides **relative measures**, need a third track to give a reference
- For absolute measuring, need to **start counting from system startup**
- Subject to **quantization** errors, **quadrature** errors, **division** error

Relative measure



- **division error**: maximum displacement between two consecutive leading/trailing edges, typically within max $\pm 25^\circ$ electrical
- the phase shift of the two channels, nominally equal to 90° electrical, is typically within max $\pm 35^\circ$ electrical (**quadrature error**)

Proprioception: **Velocity**

- **Numerical differentiation** of position
 - Backward differentiation formulas (BDFs) + filtering for smoothness
 - 1-step (Euler)

$$\dot{q}_k = \dot{q}(kT) = \frac{1}{T}(q_k - q_{k-1})$$

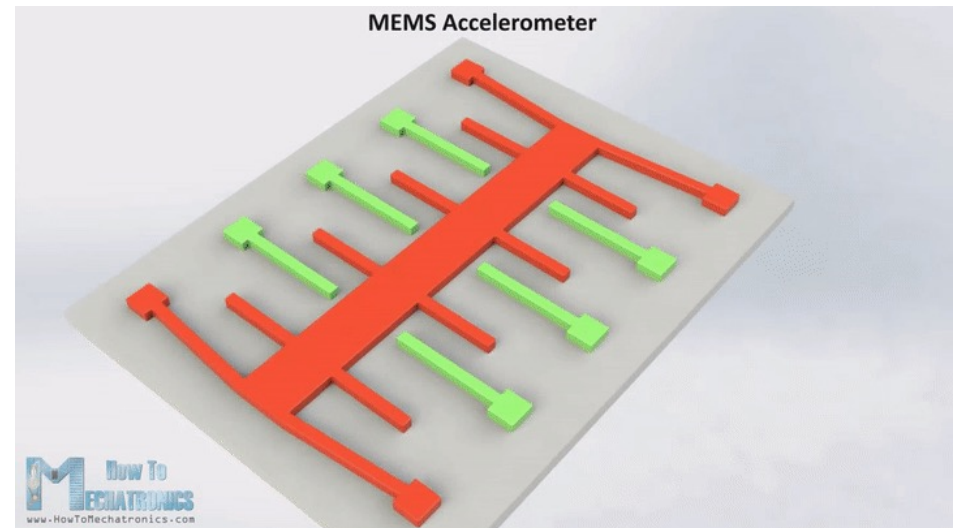
- 4-step

$$\dot{q}_k = \frac{1}{T} \left(\frac{25}{12} q_k - 4q_{k-1} + 3q_{k-2} - \frac{4}{3} q_{k-3} + \frac{1}{4} q_{k-4} \right)$$

- **Kalman Filtering**

Proprioception: Acceleration

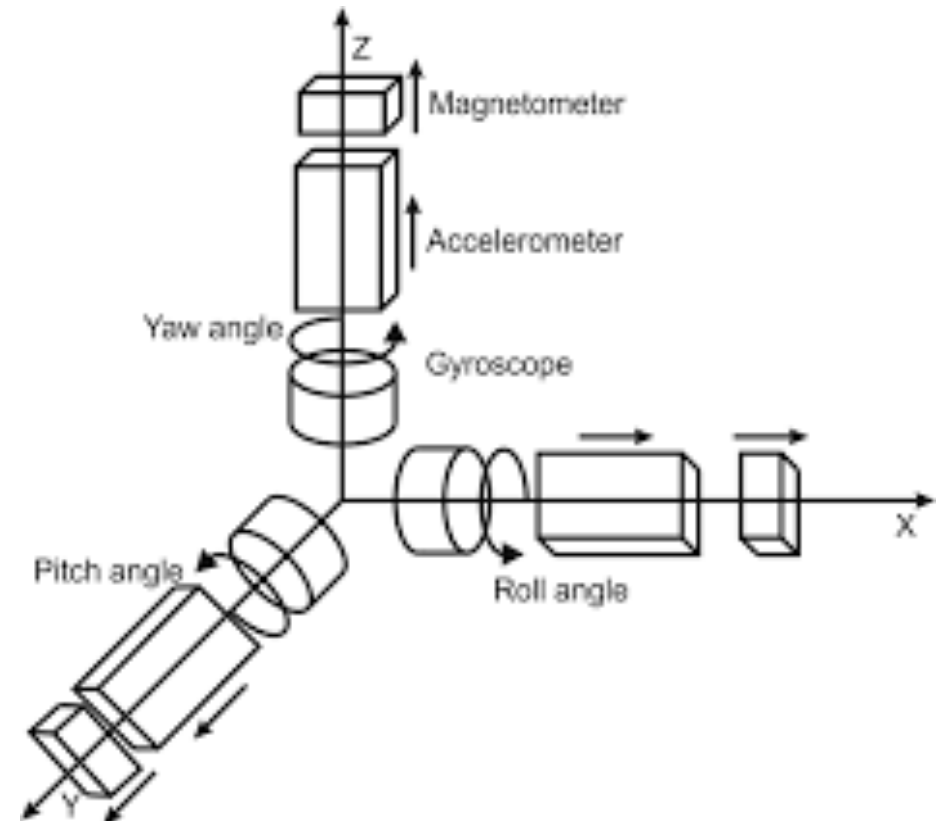
- Often through MEMS (Micro Electronic Mechanical Systems)
- Idea: measure capacity change caused by motion of a mass



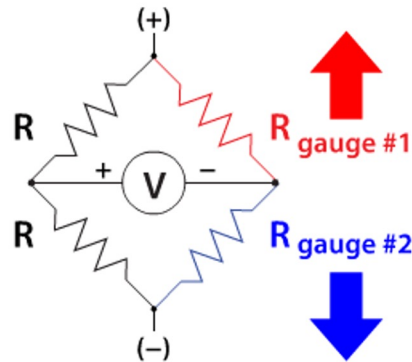
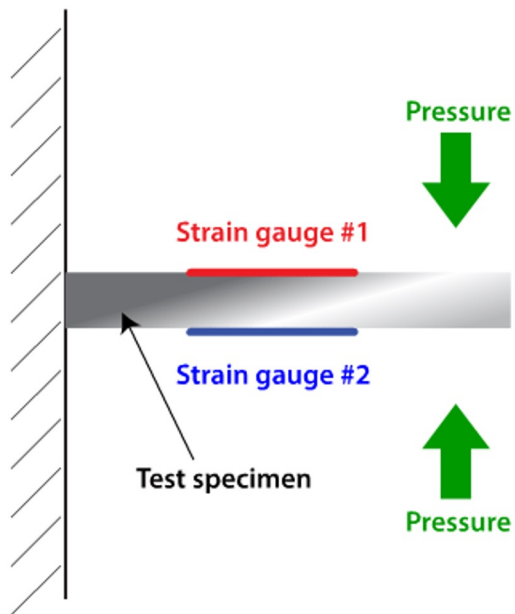
Proprioception: IMUs

- Inertial measurement units
 - Accelerometer (linear)
 - Gyroscope (angular velocity)
 - Magnetometer (direction of magnetic field)

<https://www.youtube.com/watch?v=eqZgxR6eRjo>



Exteroception: **strain gauges**



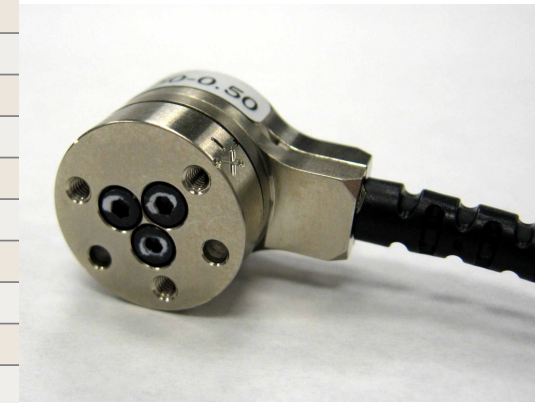
- Used to measure external linear forces
- But also deformations

Exteroception: F/T sensing

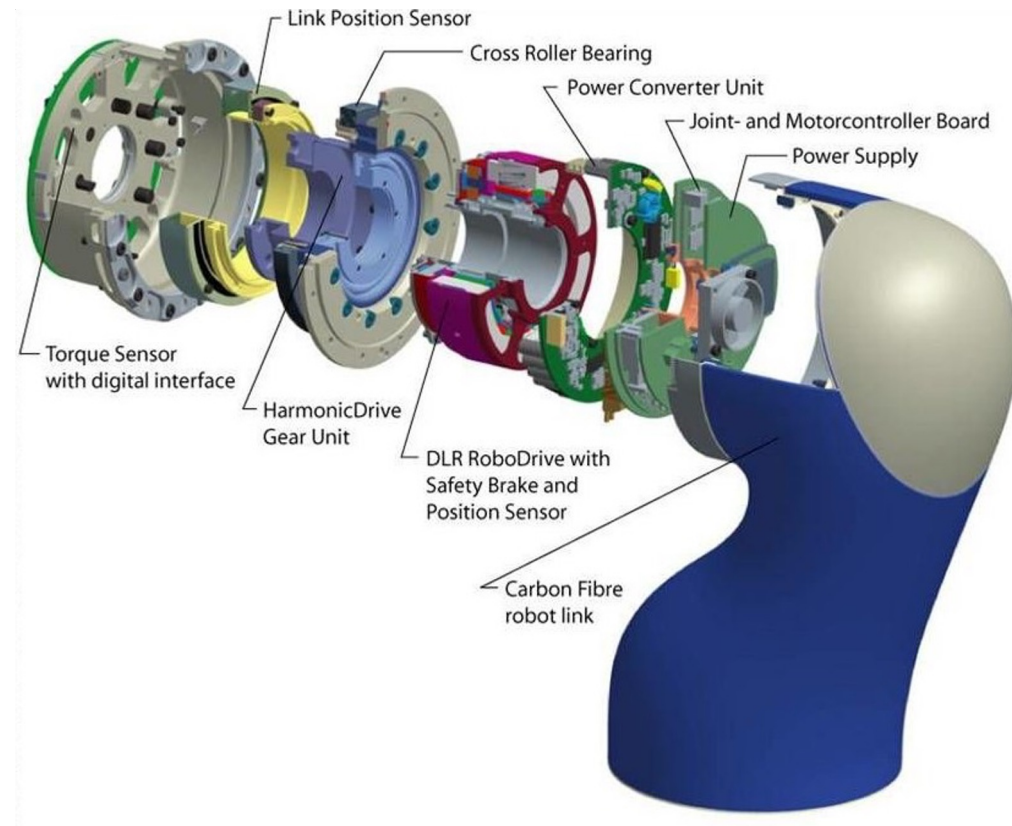


Exteroception: F/T ATI family

Model	Rated Fx,Fy	Rated Fz	Rated Tx,Ty	Weight*	Diameter*	Height*
Nano43	±36 N	±36 N	±500 Nmm	0.0387 kg	43 mm	11.5 mm
Nano17 Titanium	±32 N	±56.4 N	±200 Nmm	0.0101 kg	17 mm	14.5 mm
Nano17	±50 N	±70 N	±500 Nmm	0.00907 kg	17 mm	14.5 mm
Nano17 IP65/IP68	±50 N	±70 N	±500 Nmm	0.0408 kg	20.1 mm	22.2 mm
Omega331	±40 kN	±88 kN	±6 kNm	47 kg	330 mm	107 mm
Mini27 Titanium	±80 N	±160 N	±4 Nm	0.0334 kg	27 mm	18.2 mm
Mini40	±80 N	±240 N	±4 Nm	0.0499 kg	40 mm	12.2 mm
Mini40 IP65/IP68	±80 N	±240 N	±4 Nm	0.272 kg	53.3 mm	21.1 mm
Mini43LP	±250 N	±250 N	±3 Nm	0.0512 kg	43 mm	7.9 mm
Gamma	±130 N	±400 N	±10 Nm	0.255 kg	75.4 mm	33.3 mm
Gamma IP60	±130 N	±400 N	±10 Nm	0.467 kg	99.1 mm	39.6 mm
Gamma IP65	±130 N	±400 N	±10 Nm	1.09 kg	111 mm	52.3 mm
Gamma IP68	±130 N	±400 N	±10 Nm	1.98 kg	111 mm	52.3 mm
Axia80-M8	±150 N	±470 N	±8 Nm	0.28 kg	82 mm	25.4 mm
Mini45 Titanium	±240 N	±480 N	±12 Nm	0.0998 kg	45 mm	17.5 mm
Axia80-M20	±500 N	±900 N	±20 Nm	0.3 kg	82 mm	25.4 mm
Nano25	±250 N	±1000 N	±6 Nm	0.0634 kg	25 mm	21.6 mm
Nano25 IP65/IP68	±250 N	±1000 N	±6 Nm	0.136 kg	28 mm	27.5 mm
Mini45	±580 N	±1160 N	±20 Nm	0.0917 kg	45 mm	15.7 mm
Mini45 IP65/IP68	±580 N	±1160 N	±20 Nm	0.391 kg	57.9 mm	25.1 mm
Delta	±660 N	±1980 N	±60 Nm	0.913 kg	94.5 mm	33.3 mm



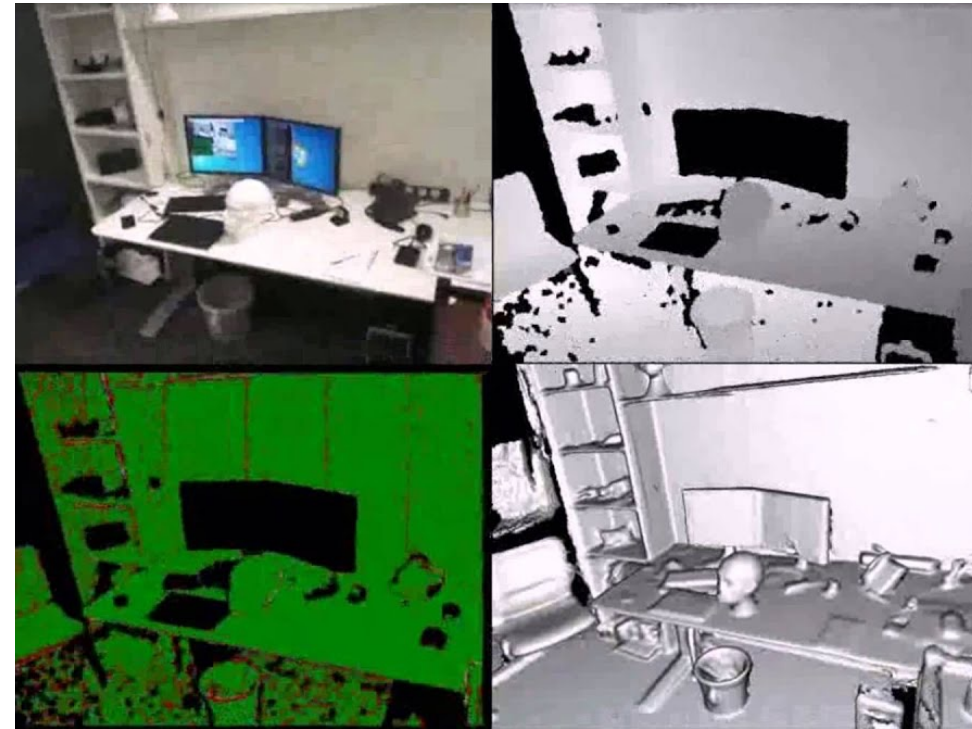
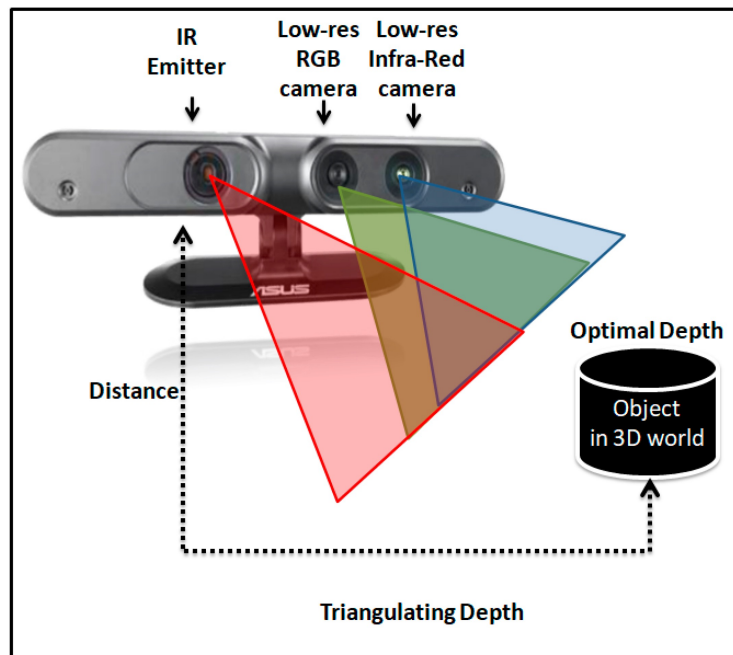
Inside a manipulator



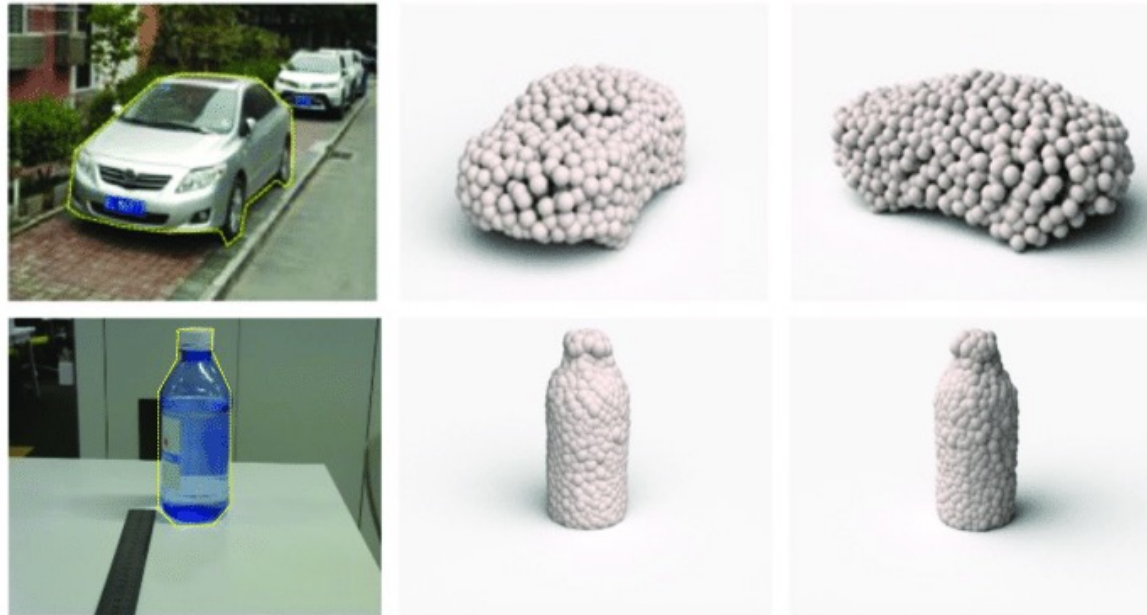
Exteroception: vision 2D



Exteroception: vision 2.5D



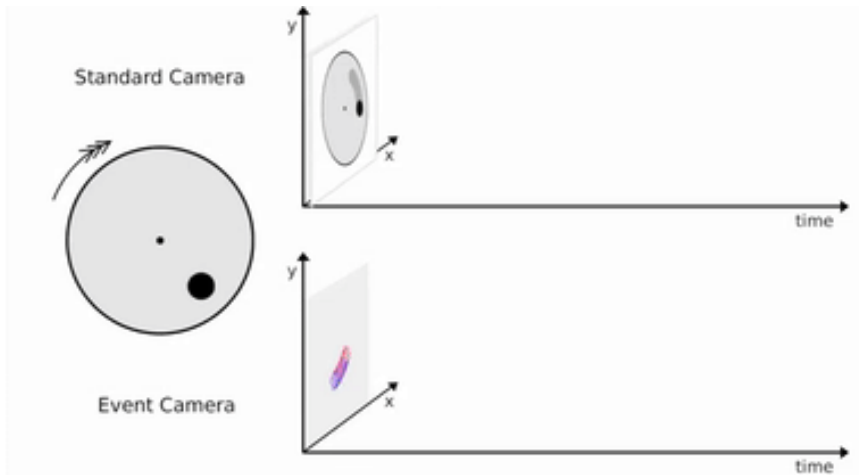
Exteroception: point clouds



Input

Reconstructed 3D point cloud

Exteroception: event cameras

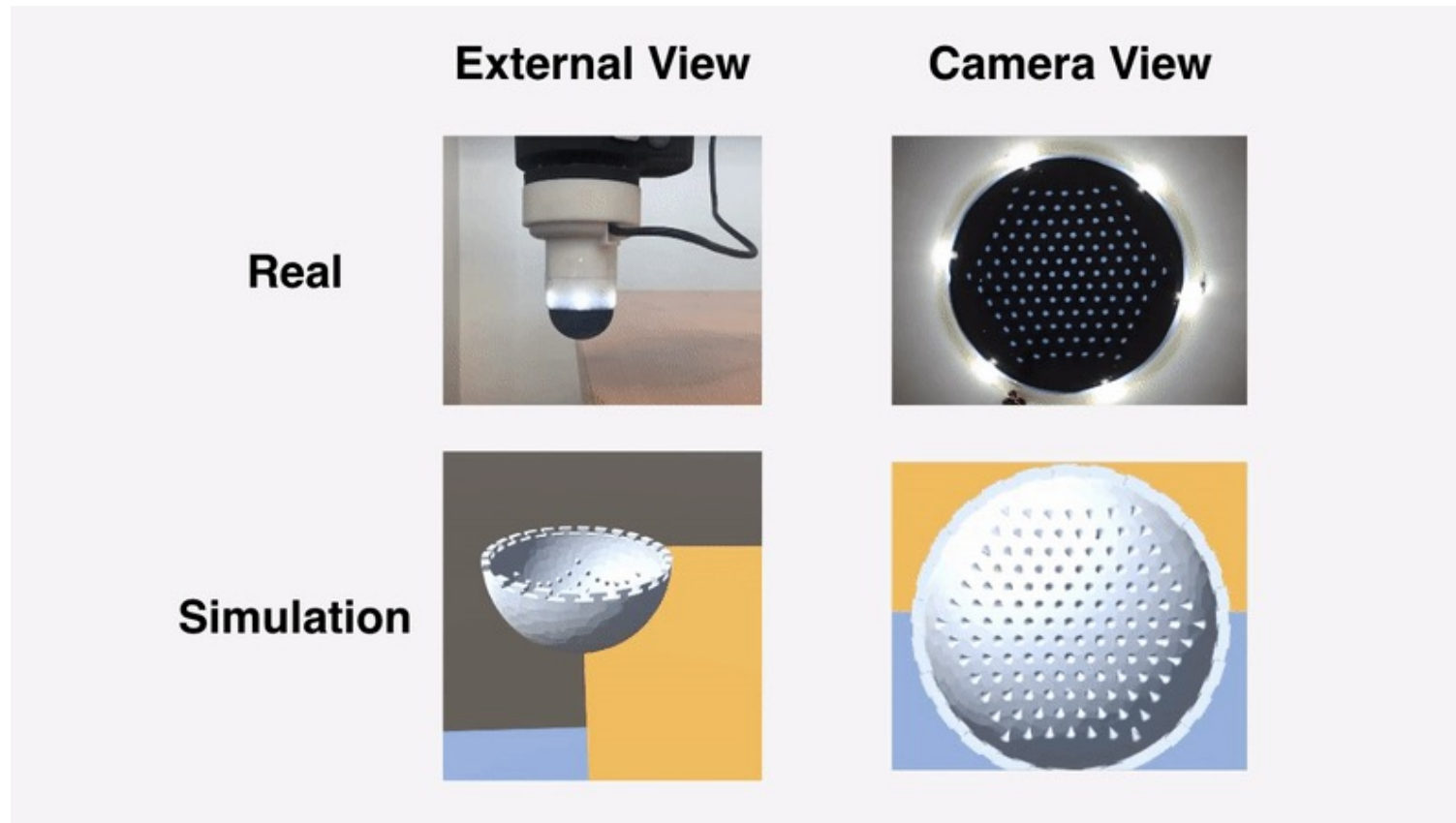


FRAME - BASED



EVENT - BASED

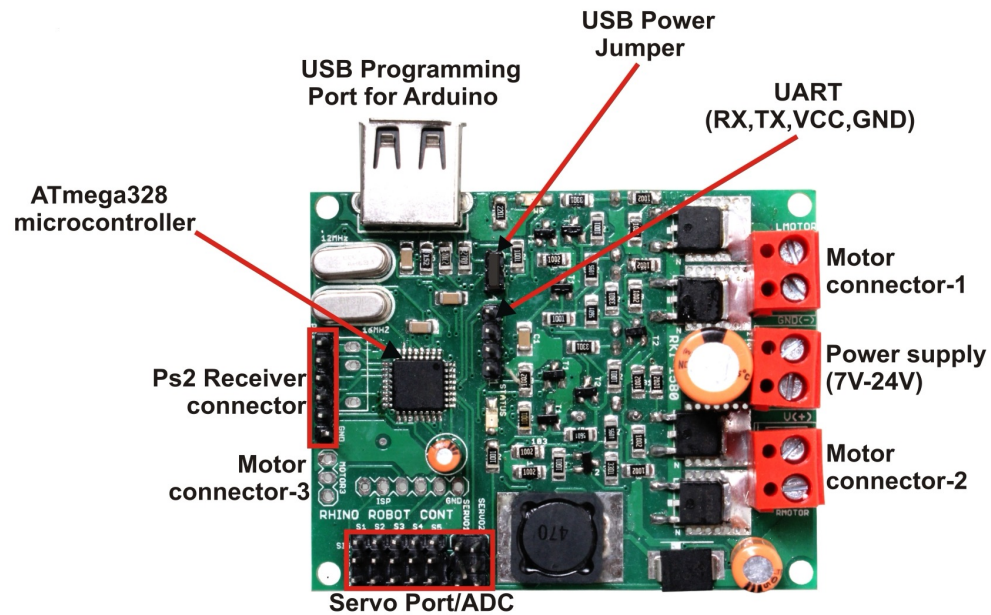
Tactile sensors



Shopping List

- **Actuators:** electro-mechanical systems which **generate forces** on links, objects, environment
- **Sensors:** electro-mechanical (but also chemical, ...) which provide a **measurable output** proportional to **changes** of a specific physical phenomenon
- **Control unit:** micro-controller, CPU, FPGA, which **regulates** the **actuators** depending on **sensors** readings

Control Units



A.Y. 2025-2026



Robot Learning

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